HUDHAIFA MUHAMMED ALASAD

DATA SCIENTIST

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PROFESSIONAL PROFILE

Data Scientist with 2 years of independent consulting experience across 6+ engagements in real estate, finance, government, and media. Develops end-to-end machine learning solutions, from data preparation and feature engineering through model evaluation, interpretation, and deployment. Delivered a price model trained on 150,000+ records (R^2 0.98) and an Arabic retrieval system with 100% Hit Rate@5 and 0.949 MRR. Current M.Sc. research focuses on Saudi dialect sentiment analysis.

EXPERIENCE

Independent Data Scientist | 2022 - Present

Remote | Gulf-market clients

- Led 6+ consulting engagements from problem definition and data preparation to model development, deployment, and presentation of actionable findings to senior stakeholders.
- Developed real estate price-forecasting pipelines using location, area, and pricing-history features; identified 3 high-ROI zones that informed acquisitions and contributed to approximately 15% growth in a client portfolio.
- Built RFM-based customer segments and churn-risk models to prioritize retention spending across customer groups.
- Created SQL ETL workflows and automated KPI reporting for an HR system, covering payroll, attendance, and satisfaction data for 500+ employees.

PROJECTS

Used Car Market Price Prediction | Predictive Modeling

- Engineered features and tuned XGBoost models on 150,000+ records; achieved R^2 0.98, interpreted predictions with SHAP, and deployed real-time inference in Streamlit.

Arabic Multi-Tool RAG Agent | Arabic NLP / Information Retrieval

- Designed a retrieval pipeline with multilingual-e5-large embeddings, ChromaDB, and LangChain; achieved 100% Hit Rate@5 and 0.949 MRR on Arabic regulatory documents.

Food Waste Forecasting & Severity Classification | Machine Learning

- Combined quantity forecasting (R^2 0.96) and severity classification (85% accuracy) with a 3×3 decision matrix to generate automated recommendations.

Arabic Dialect Sentiment Analysis | M.Sc. Research in Progress

- Fine-tuning AraBERT, MARBERT, and CAMeLBERT on Saudi product-review datasets to study dialect-specific sentiment classification and cross-dialect transfer.

TECHNICAL SKILLS

Machine Learning: Scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, regression, classification, clustering, model evaluation, SHAP

Data & Statistics: Python, Pandas, NumPy, SQL, feature engineering, EDA, hypothesis testing, statistical modeling, ETL

NLP: Hugging Face Transformers, AraBERT, RAG, LangChain, ChromaDB

Deployment & MLOps: Streamlit, Flask, Docker, Git, MLflow

EDUCATION

M.Sc. Computer Science (In Progress) | 2024 - Present

Sana'a University, Yemen | Thesis: Transformer-based sentiment analysis of Saudi dialect consumer reviews

B.Sc. Information Technology | 2019 - 2023

Sana'a University, Yemen

CERTIFICATIONS & TRAINING

AWS Machine Learning Engineer - Associate course (2026)

Machine Learning Specialization, Stanford University (2024)

IBM Data Science Professional Specialization (2023)

LANGUAGES

Arabic: Native | English: Upper-Intermediate (B2)